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LMAPR2014 : Physics of Functional Materials

Gemstones and Lasers

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Contents

1 Introduction **2**

2 Fundamentals of Light-Matter interaction **2**
2.1 The LASER Effect: A Qualitative Overview . . . 2
2.2 Einstein coefficients . . . 3
2.3 Hamiltonian with electromagnetic field . . . 4
2.4 Two-level system . . . 5
2.5 Multilevel Systems and Optical Pumping . . . 7
2.6 Photoluminescent Mechanism . . . 9

3 Electronic structure of isolated atoms and origin of color **10**
3.1 Electronic structure of isolated atoms . . . 10
3.2 Spin-Orbit coupling . . . 11
3.3 Crystal field theory . . . 12
3.3.1 Cr^{3+} in Octahedral Field (Ruby) . . . 13
3.3.2 Ti^{3+} in octahedral field (Sapphire) . . . 14

4 Gemstones: description and properties **15**
4.1 Structure and mechanical properties . . . 16
4.1.1 Sapphire and ruby . . . 16
4.1.2 Alexandrite . . . 16
4.2 Optical properties and the Alexandrite effect . . . 17
4.3 Synthetic gemstones and fabrication processes . . . 18
4.3.1 Czochralski method . . . 18
4.3.2 Verneuil furnace . . . 19

5 Gemstones-based LASERs **19**
5.1 Ruby LASER . . . 19
5.2 Ti:Sapphire LASER . . . 20
5.3 Alexandrite LASER . . . 21

6 Market and prospectives **21**
6.1 Market analysis . . . 21
6.2 Future Prospectives: Miniaturization and Strategic Sovereignty . . . 23

7 Conclusion and Outlook **24**

1 Introduction

In 1917, Albert Einstein published the paper "On the Quantum Theory of Radiation", which constitutes the foundations of light-matter interaction, based upon probability coefficients (Einstein coefficients) for the absorption, spontaneous and stimulated emission of electromagnetic radiation. The experimental confirmation of these phenomena in the following decades allowed for numerous research groups to try and create the amplification of electromagnetic radiation. Particularly, in 1953, Charles H. Townes, James P. Gordon and Herbert J. Zeiger produced the first microwave amplifier (commonly called MASER), which works on the same principle as a laser but amplifies microwave radiation, whereas the lasers we know of amplify infrared or visible radiation[1].

This success in the microwave spectrum spurred an intense scientific race to create an optical equivalent. This race was won in 1960 by Theodore Maiman at Hughes Research Laboratories. Maiman demonstrated the first working laser, and his gain medium was a synthetic ruby crystal—a gemstone[2]. This fundamental link between gemstones and laser technology, established at its very inception, forms the basis of our report.

This report, realized in the frame of the LMAPR2014-Physics of Functional Materials lecture taught at UCLouvain, aims to answer 3 main questions :

- What is the origin of colour in gemstones and is it tunable at will ?
- What is the working principle of a LASER ?
- Why are some of the most powerful LASERs based on gemstones ?

We will start by introducing the necessary notions of physics, which include energy levels, crystal field theory and light-matter interaction. We will then be able to properly explain the working principle of a laser and the photoluminescence phenomenon in doped insulators, as well as gemstone properties and their use in laser devices. Finally, an overview of the current market regarding gemstones and lasers will be given, as well as future perspectives and areas of improvement.

2 Fundamentals of Light-Matter interaction

To understand the operation of lasers, geometric optics alone is not sufficient : it is necessary to introduce the principles of the quantum interaction between light and matter. A complete description would require the quantization of the electromagnetic field, but for the purposes of this work, a semiclassical approach (in which light is treated as a classical, continuous field and matter as a quantum system) is sufficient. This perspective makes it possible to grasp the fundamental mechanisms underlying the optical phenomena that enable laser operation.

2.1 The LASER Effect: A Qualitative Overview

We will first give an overview of the LASER effect and conditions of realization, before diving deeper into the theoretical physics that underlie this phenomenon. The first thing to assess is the LASER acronym itself: Light Amplification by Stimulated Emission of Radiation. Stimulated emission is the process through which a sample can amplify a beam of light directed towards it by undergoing an atomic transition from an excited state to a lower energy state, creating a new photon as a result, which happens to have the same phase and polarization as the incident one.

However, this process is in direct competition with absorption. In any normal material at thermal equilibrium, the vast majority of atoms are in the ground state, not an excited state. Since the rates of absorption and stimulated emission are respectively proportional to the population of the ground state and excited state, this means that a photon traveling through the material is overwhelmingly

more likely to be absorbed by a ground-state atom than it is to stimulate emission from an excited atom.

Therefore, under normal conditions, any sample of material will attenuate (weaken) light, not amplify it. To achieve amplification and build a laser, we must first find a way to tip the scales and reverse this natural situation. This process is called a **population inversion**, because the aim is to populate the excited state *more* than the ground state, thus giving a larger probability of stimulated emission than absorption[3]. The material capable of sustaining this population inversion is formally called the **gain medium** (or active medium). It acts as the physical source of optical amplification. In the specific context of this report, the gain medium is a solid-state crystal (a gemstone) doped with transition metal ions, for reasons that will be deeply explored in the following.

The last element required to build up a LASER beam is an **optical cavity**. Once amplification is possible, the photon beam must be confined inside a cavity that contains this gain medium. This is typically achieved thanks to 2 mirrors placed on each side of the cavity. The beam will travel back and forth between the mirrors, and trigger more and more stimulated emissions during each travel. One of those mirrors has to be semi-transparent to let a small fraction of the beam escape through it. A schematic representation of a LASER cavity is given in figure 1.

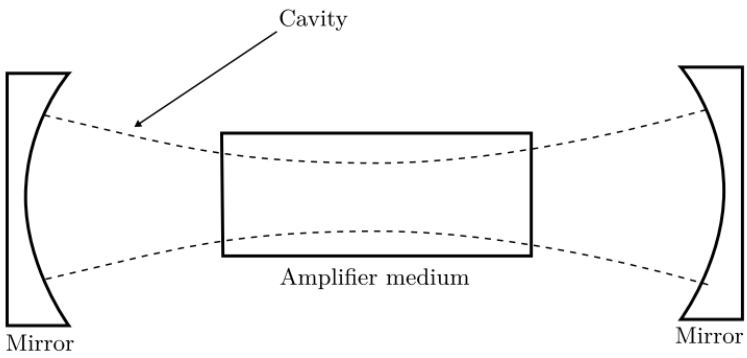


Figure 1: Confinement of a LASER beam[3].

While the macroscopic components of a laser—mirrors, gain medium, and the resulting intense beam—are conceptually elegant, a purely qualitative description is insufficient to explain why amplification occurs in some materials but not others. To truly understand the conditions required for a laser to function, we must move beyond this physical description and investigate the statistical and quantum mechanical laws governing atomic transitions. The following sections will therefore establish the mathematical framework of light-matter interaction, starting with Einstein's rate equations.

2.2 Einstein coefficients

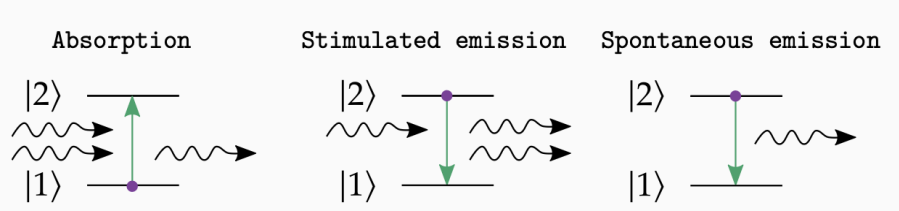


Figure 2: Three elementary processes in semi-classical light matter interaction[4].

In a closed system at the thermodynamical equilibrium, constituted of N atoms either in energy states 1 or 2 (ground state/excited state), Einstein showed that the following three processes govern the interaction between matter and broadband radiation :

- **Absorption:** A system initially in the ground state can absorb a photon of frequency ν, thus transitioning to the higher energy level.
- **Spontaneous emission:** A system in the excited state may spontaneously return to the lower level, emitting a photon of frequency ν, provided that the transition between E_2 and E_1 is radiative.
- **Stimulated emission:** In this case, the system returns from the excited to the ground state, but the process is triggered by an incident photon. The emitted photon has the same characteristics—frequency, phase, and direction—as the photon that induced the transition.

Before introducing the equations that describe the time evolution of the populations in the two levels, it is useful to clarify the physical meaning of the Einstein coefficients. These coefficients provide a phenomenological description of all the processes through which an atomic system can exchange energy with a radiation field: absorption, spontaneous emission, and stimulated emission. The combined action of these three mechanisms determines how the level populations evolve in the presence of electromagnetic radiation. The following equations therefore express the overall balance of these contributions :[3][4]

$$\frac{dN_2}{dt} = -A_{21}N_2 + B_{12}\,\rho(\omega_{12})\,N_1 - B_{21}\,\rho(\omega_{12})\,N_2, \tag{1}$$

$$\frac{dN_1}{dt} = A_{21}N_2 - B_{12}\,\rho(\omega_{12})\,N_1 + B_{21}\,\rho(\omega_{12})\,N_2. \tag{2}$$

where:

- A_{21} is the Einstein coefficient for spontaneous emission,
- B_{12} is the Einstein coefficient for absorption,
- B_{21} is the Einstein coefficient for stimulated emission,
- $\rho(\omega)$ is the spectral energy density of the radiation at frequency ω,
- N_1 represents the population in state 1,
- N_2 represents the population in state 2.

The Einstein coefficients provide a robust phenomenological framework for describing the balance of rates in a closed system, yet they do not explain the microscopic origin of these transitions. To calculate the values of these coefficients from first principles and understand how an electromagnetic wave physically perturbs the atomic wavefunction, we must turn to a rigorous quantum mechanical treatment. In the following section, we introduce the full Hamiltonian of the system, explicitly coupling the atom to the external electromagnetic field.

2.3 Hamiltonian with electromagnetic field

In the presence of an external electromagnetic field, introduced via the transverse vector potential $\mathbf{A}_\perp(\mathbf{r}, t)$ in the Coulomb gauge, the Hamiltonian takes the form:

$$\hat{H}(t) = \frac{1}{2m}[\hat{\mathbf{p}} - q\,\mathbf{A}_\perp(\hat{\mathbf{r}}, t)]^2 + V(\hat{\mathbf{r}}) = \hat{H}_0 + \hat{W}(\hat{\mathbf{r}}, t), \tag{3}$$

where $\hat{H}_0$ is the hamiltonian of an isolated atom, described in the nonrelativistic limit:

$$\hat{H}_0(\hat{\mathbf{r}}) = \frac{\hat{\mathbf{p}}^{\,2}}{2m} - \frac{q^2}{4\pi\varepsilon_0\,\hat{r}} = \frac{\hat{\mathbf{p}}^{\,2}}{2m} + \hat{V}(\hat{\mathbf{r}}). \tag{4}$$

and $\hat{W}(\hat{\mathbf{r}}, t)$ is the perturbation caused by the electromagnetic field.

If the wavelength of the electromagnetic field is much larger than the size of the atom ($\lambda \gg a_0$), the vector potential can be approximated as constant over the atomic region. This is equivalent to expanding the exponential terms and retaining only the first term of the expansion, which leads to the dipole approximation. For this reason we denote the interaction potential with the subscript E_1. The quadratic term does not induce transitions:

$$\hat{W}_{E1}(\mathbf{r}_0, t) = -\frac{q}{m}\, \mathbf{A}_\perp(\mathbf{r}_0, t) \cdot \hat{\mathbf{p}}. \tag{5}$$

or, expanding the vector potential:

$$\hat{W}_{E1}(\mathbf{r}_0, t) = -\frac{qA_0}{m}\, \boldsymbol{\epsilon} \cdot \hat{\mathbf{p}}\, \cos(\omega t - \varphi). \tag{6}$$

where $\varphi = \phi + \mathbf{k} \cdot \mathbf{r}_0$.

A form that is more convenient for calculations is obtained by a gauge transformation, yielding:

$$W_{E1}(\mathbf{r}_0, t) = -E_0\, \mathbf{D} \cdot \boldsymbol{\epsilon}\, \cos(\omega t - \varphi). \tag{7}$$

where $\hat{\mathbf{D}} = q\,(\mathbf{r} - \mathbf{r}_0)$ is the electronic dipole operator. Having established this Hamiltonian, we can now apply it to a two-level system to derive the influence of light radiation on the populations of the levels. [4] [5]

2.4 Two-level system

The simplest way to describe the dynamics of light–matter interaction is to consider a quantum system with only two energy levels. Although atoms and molecules possess an infinite number of levels, the two-level approximation is physically valid when the photon energy, $\hbar\omega$, is close to the energy difference between two non-degenerate levels of the system, $\Delta E = E_2 - E_1$. Under this near-resonance condition, the interaction dynamics is dominated by these two levels, while the contribution from all others may be neglected. The system therefore interacts resonantly with the electromagnetic field. The dynamics of the system subject to the electromagnetic field are described by the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \left[\, \hat{H}_0 + \hat{W}_{E1}(\mathbf{r}_0, t) \,\right] \Psi(\mathbf{r}, t). \tag{8}$$

Recalling that any time-dependent state of a two-level system can be written as a combination of the eigenstates $|1\rangle$ and $|2\rangle$:

$$\Psi(\mathbf{r}, t) = c_1(t)\, e^{-iE_1 t/\hbar}\, |1\rangle + c_2(t)\, e^{-iE_2 t/\hbar}\, |2\rangle\,. \tag{9}$$

In the basis set $\{|1\rangle\,, |2\rangle\}$, the Hamiltonian is:

$$\hat{H}(t) = \begin{pmatrix} E_1 & -E_0\, \langle 1|\mathbf{D}\cdot\boldsymbol{\epsilon}|2\rangle \cos(\omega t - \varphi) \\ -E_0\, \langle 2|\mathbf{D}\cdot\boldsymbol{\epsilon}|1\rangle \cos(\omega t - \varphi) & E_2 \end{pmatrix}. \tag{10}$$

Defining the *Rabi frequency*

$$\Omega = -\frac{E_0}{\hbar}\, \langle 1|\mathbf{D}\cdot\boldsymbol{\epsilon}|2\rangle$$

and the atomic transition frequency $\omega_{21} = (E_2 - E_1)/\hbar$, and introducing the rotating-wave approximation (neglecting rapidly oscillating terms), the coupled equations become:

$$i\dot{c}_1(t) = \frac{\Omega}{2}\, e^{\,i(\omega - \omega_{21})t - i\varphi}\, c_2(t), \tag{11}$$

$$i\dot{c}_2(t) = \frac{\Omega^*}{2}\, e^{-i(\omega - \omega_{21})t + i\varphi}\, c_1(t). \tag{12}$$

It is useful to introduce the *detuning* $\delta = \omega - \omega_{21}$, which measures the deviation from resonance.

It is convenient to introduce a change of variables that removes the explicit time dependence from the off-diagonal elements of the effective Hamiltonian. We define:

$$c_1(t) = \tilde{c}_1(t)\, e^{+i\delta t/2}, \tag{13}$$

$$c_2(t) = \tilde{c}_2(t)\, e^{-i\delta t/2}, \tag{14}$$

Substituting (13)–(14) into the equations for $c_1(t)$ and $c_2(t)$ and imposing the initial conditions

$$c_1(0) = 1, \qquad c_2(0) = 0,$$

we obtain:

$$\tilde{c}_1(t) = \cos\left(\frac{\tilde{\Omega} t}{2}\right) - i\,\frac{\delta}{\tilde{\Omega}}\,\sin\left(\frac{\tilde{\Omega} t}{2}\right), \tag{15}$$

$$\tilde{c}_2(t) = -i\,\frac{\Omega\, e^{-i\varphi}}{\tilde{\Omega}}\,\sin\left(\frac{\tilde{\Omega} t}{2}\right), \tag{16}$$

where we introduced the *generalized Rabi frequency*:

$$\tilde{\Omega} = \sqrt{\Omega^2 + \delta^2}. \tag{17}$$

From the solution (16), it follows that:

$$|c_2(t)|^2 = |\tilde{c}_2(t)|^2 = \frac{\Omega^2}{\Omega^2 + \delta^2}\,\sin^2\left(\frac{\tilde{\Omega} t}{2}\right), \tag{18}$$

while:

$$|c_1(t)|^2 = 1 - |c_2(t)|^2. \tag{19}$$

The probability of finding the system in the excited state, denoted $P_2(t)$, is therefore:

$$P_2(t) = \frac{\Omega^2}{\Omega^2 + \delta^2}\,\sin^2\left(\frac{\tilde{\Omega} t}{2}\right). \tag{20}$$

This expression shows that the population of the excited state oscillates in time with frequency $\tilde{\Omega}$: these periodic oscillations are the *Rabi oscillations* 3. The maximum amplitude of the oscillations depends on both the field amplitude (through Ω) and the detuning δ: in resonance ($\delta = 0$) one has

$$P_2^{\max} = 1,$$

while off-resonance the amplitude is reduced by the factor $\Omega^2/(\Omega^2 + \delta^2)$ [4].

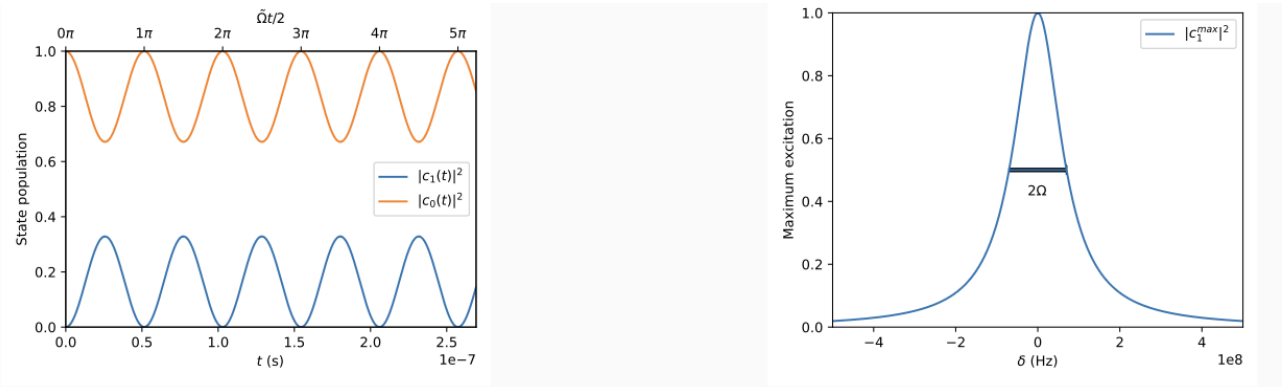


Figure 3: Rabi oscillations for $\Omega = 7.0 \times 10^7\,\mathrm{s}^{-1}$ and $\delta = 1.0 \times 10^8\,\mathrm{s}^{-1}$(left) and Maximum excitation probability for $\Omega = 7.0 \times 10^7\,\mathrm{s}^{-1}$ (right)[4].

Let us connect our calculations to Einstein's treatment of the interaction between matter and *broadband* radiation. In the frequency interval $[\omega,\, \omega + d\omega]$, the radiation has an energy $\rho(\omega)\, d\omega = \varepsilon_0 E_0^2(\omega)/2$. Integration of Eq. 18 over the spectral distribution of the radiation yields:

$$|c_2(t)|^2 = \int_{\Delta\omega} \frac{|E_0 \langle 1|\mathbf{D}\cdot\boldsymbol{\epsilon}|2\rangle|^2}{4\hbar^2} t^2 \frac{\sin^2\left(\tilde{\Omega}\, t/2\right)}{\left(\tilde{\Omega}\, t/2\right)^2}\, d\omega. \tag{21}$$

From this expression we obtain the remarkable result:

$$B_{21} = \frac{\pi\, |\langle 1|\mathbf{D}\cdot\boldsymbol{\epsilon}|2\rangle|^2}{\varepsilon_0 \hbar^2}. \tag{22}$$

Thermodynamical arguments further imply:

$$B_{21} = \frac{g_1}{g_2}\, B_{12}. \tag{23}$$

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} = \frac{\hbar\omega^3}{\pi^2 c^3}. \tag{24}$$

The above results highlight a fundamental limitation of a two-level system. Since absorption and stimulated emission occur with equal probability (when $g_1 = g_2$), the radiation field cannot create or sustain a population inversion. As soon as the excited-state population increases, emission from the excited state compensates for it, driving the system back toward thermal equilibrium. The maximum achievable excitation is therefore $N_2 = N_1$, and the condition $N_2 > N_1$ required for optical gain can never be reached.

For this reason, practical laser operation cannot rely on a two-level configuration. A minimum of three energy levels is required so that fast decay channels and metastable states enable an asymmetric population flow. Only in such multilevel schemes can the population accumulate in an excited state long enough to achieve and maintain a true population inversion[3][4].

2.5 Multilevel Systems and Optical Pumping

To overcome the limitation of the two-level system and achieve the population inversion condition ($N_{excited} > N_{ground}$), we must introduce intermediate energy levels. This allows us to separate the excitation step (pumping) from the emission step. The process of actively populating the excited states is called **pumping**. For the gemstones discussed in this report, we rely on **optical pumping**, which uses an intense light source to excite the medium. Lasers based on this principle generally fall into two categories: 3-level and 4-level systems (Fig. 4). Compared to the two-level case, such systems exhibit additional phenomena, such as multiple couplings, two-photon transitions, and differentiated light shifts.

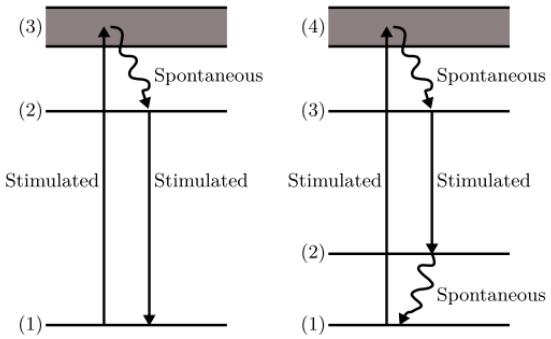


Figure 4: Schematic of 3-level and 4-level lasers[3].

Optical pumping is a process by which light is used to modify the population distribution among the energy levels of a medium. The selective absorption of photons can transfer atoms from a set of initially populated quantum states to different states in a controlled manner. This mechanism

underlies the operation of many laser systems, including gemstone lasers, where it is necessary to efficiently populate metastable excited levels.

A photon possesses energy $E = \hbar\omega$, momentum $p = \hbar k$, and total angular momentum $j = 1$. Its possible polarizations correspond to angular momentum projections $m_j = \pm 1$ (circular polarizations σ_+ and σ_-) or to their linear combination (linearly polarized light π). The absorption of a photon by an atom must conserve both energy and angular momentum, imposing selection rules on electric dipole transitions:

$$\Delta m = 0, \pm 1, \tag{25}$$

depending on the polarization of the incident light. The use of σ_+, σ_-, or π light thus allows one to select which magnetic sublevels m of a given ground state can be excited [4] [6] [7].
Optical pumping is not simply a redistribution among magnetic sublevels: it can be engineered to drive the atomic population out of thermal equilibrium and into metastable excited states, which is precisely the mechanism exploited in LASERs.

Returning to the discussion of three-level systems, let us now consider the case in which only one laser field is present ($\Omega_2 = 0$), resonant with the $1 \to 2$ transition ($\delta_2 = 0$).

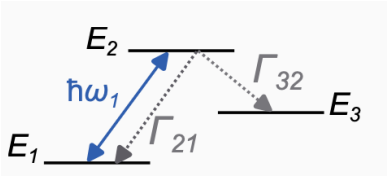


Figure 5: Three-level system driven by a single laser field[4]. The quantities Γ_{21} and Γ_{32} denote the spontaneous emission rates.

As shown in Fig. 5, under steady-state conditions the entire population is transferred to level 3. By imposing suitable constraints on the lifetimes of the states and on the couplings between them, it becomes evident why a three-level system is an excellent candidate for implementing a LASER scheme.

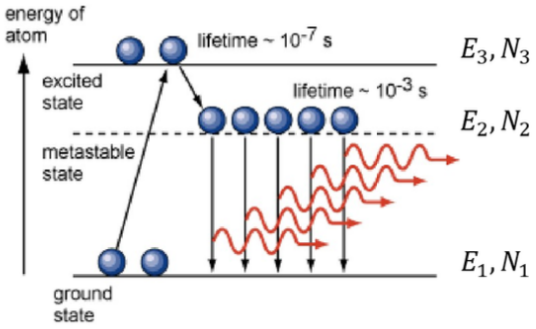


Figure 6: Example of population inversion for a three-level LASER. Population inversion is reached when $N_2 > N_1$. The lifetime of the excited state is so short that we can assume $N_3 = 0$[8].

Figure 6 shows the classical three-level scheme found in many solid-state lasers:

- The level E_3 is excited by the pumping radiation.
- The lifetime of E_3 is very short (on the order of 10^{-7} s).
- The excited level rapidly decays to an intermediate level E_2.
- The level E_2 is metastable, with a much longer lifetime (typically on the order of 10^{-3} s).

Because of this difference in lifetimes, the population that reaches E_2 tends to accumulate. When the population in the metastable level exceeds that of the ground level E_1, a *population inversion* is achieved. The scheme shown in the figures illustrates this clearly: the rapid decay $E_3 \to E_2$ and the slow decay $E_2 \to E_1$ make the intermediate level the population reservoir.

In a 4-level laser, the electrons are pumped from level 1 to 4, then quickly and spontaneously relax to level 3, where the lifetime is quite high compared to $4 \to 3$ and $2 \to 1$ transitions. Then, by stimulated emission, they all make a transition from 3 to 2, from which they quickly relax to 1. This system is more widely used than the 3-level laser, for which the population inversion condition is $N_2 > N_1$, since in a 4-level laser, level 2 has a quite small population in the beginning and the population inversion condition is $N_3 > N_2$, which is much easier to achieve.

Other pumping methods exist, such as carrier injection pumping, chemical pumping or electronic pumping, but they are adapted for gas lasers, dye lasers or semiconductor lasers. For the sake of our project, we will focus on solid-state lasers that use optical pumping.

2.6 Photoluminescent Mechanism

We have just established that 4-level systems are ideal for maintaining population inversion. In solid-state insulators, the physical mechanism that creates this favorable energy structure is photoluminescence.

In an insulator, when the material absorbs a photon with energy exceeding the band gap energy, an electron is excited from the valence band to the conduction band. This creates a hole in the valence band, resulting in an electron-hole pair. The excited electron will dissipate its excess energy through electron-phonon or electron-electron collisions, thereby relaxing down to the lowest level of the conduction band. Similarly, the hole will rise to the highest level of the valence band. Subsequently, a recombination occurs between this conduction band electron and this valence band hole. This recombination can take place in two distinct ways: either as a radiative recombination, where energy is dissipated through the emission of a photon, or as a non-radiative process, where energy is dissipated via phonon emission, the Auger effect, or by transferring energy to impurities. Figure 7 represents this effect.

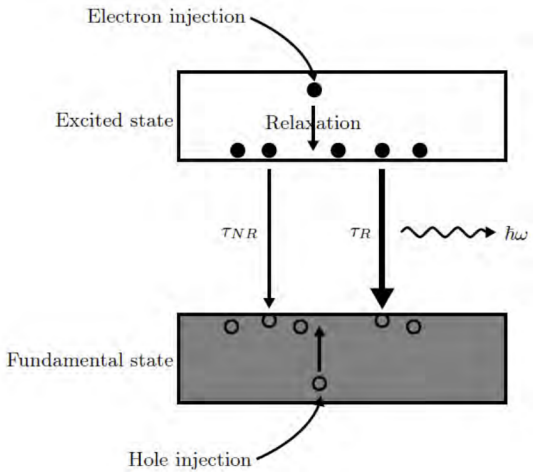


Figure 7: Luminescence mechanism [3].

There is, therefore, a competition between these two types of recombination. However, the process of interest here is evidently radiative recombination, as it is the one responsible for light production. To determine which recombination mechanism dominates, one must examine their respective lifetimes. Let τ_R denote the radiative recombination lifetime and τ_{NR} the non-radiative recombination lifetime. A high-quality luminescent material is thus characterized by the condition $\tau_R \ll \tau_{NR}$.

In the luminescence principle described above, the emitted photon consistently possesses lower energy than the absorbed photon. This can be explained by the fact that photon absorption rarely occurs into the lowest vibronic level of the excited state. The energy associated with the absorbed

photon is given by:

$$\hbar\omega_a = \epsilon_1 + (n_1 + \frac{1}{2})\hbar\Omega_1 - (\epsilon_0 + \frac{\hbar\Omega_0}{2}) = \hbar\omega_0 + n_1\hbar\Omega_1 \tag{26}$$

However, prior to recombination, as previously mentioned, non-radiative relaxation of the electron to the lowest vibronic level of the excited state occurs. Electronic recombination then proceeds from this lowest level to a vibronic level of the ground state. The emitted photon will thus have an energy corresponding to:

$$\hbar\omega_e = \epsilon_0 + (\frac{\hbar\Omega_0}{2}) - (\epsilon_1 + (n_1 + \frac{1}{2}))\hbar\Omega_1 = \hbar\omega_0 - n_1\hbar\Omega_1 \tag{27}$$

Figure 8 illustrates this phenomenon, known as the Stokes shift.

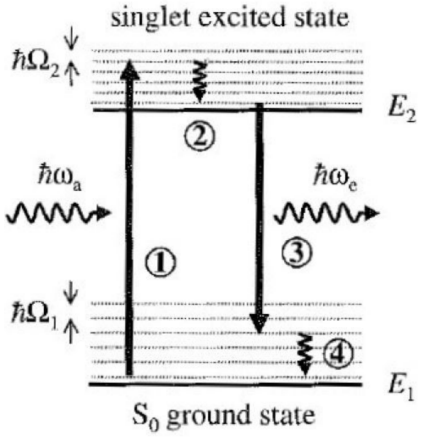


Figure 8: Stokes shift effect [9].

The Stokes shift greatly facilitates the realization of a four-level system. As a reminder, in a four-level system, the terminal level must be naturally near-empty and must depopulate very rapidly. However, in the absence of a Stokes shift, emission and absorption would correspond to similar energies; consequently, an emitted photon could be immediately reabsorbed, causing the terminal level to become populated. Furthermore, since the terminal level corresponds to an excited vibronic level of the ground state, relaxation is extremely fast. Therefore, because this terminal level remains virtually unpopulated, even a small population accumulation in level 3 is sufficient to achieve population inversion.

3 Electronic structure of isolated atoms and origin of color

To understand how multilevel laser systems and optical pumping operate in real crystals, it is essential to examine the microscopic origin of the energy levels involved. While the previous section described the population dynamics between generic laser levels, the actual structure of these levels in solid-state lasers is determined by the crystal field theory. This theory not only defines the positions of the pumping and lasing transitions but also govern selection rules, degeneracies and nonradiative relaxation pathways.

Before introducing crystal-field theory, we begin by reviewing the electronic structure of isolated atoms in the first section, followed by a discussion of spin–orbit coupling. This preliminary framework is necessary because spin–orbit interactions play a central role in determining the energy-level structure of chromium in ruby, and thus must be understood prior to examining the effects of the crystal field.

3.1 Electronic structure of isolated atoms

In an isolated atom, electrons are confined around the nucleus by the Coulomb attraction. More specifically, the behavior of the electrons is governed by the time-independent Schrödinger equation. The solutions to this equation are atomic orbitals, each described by a wavefunction. These orbitals give the probability distribution of finding the electron in space. Each orbital has a characteristic shape : s orbitals are spherical, p orbitals are dumbbell-shaped, d orbitals are toroidal, and so on. Those wavefunctions describe probability clouds, not classical trajectories. The order of filling these orbitals is dictated by the Aufbau principle, according to which electrons occupy the lowest available energy orbitals first. This order can be predicted by the combination of the principal quantum number

n and azimuthal quantum number ℓ for each orbital. Hund's rule also tells us that, for orbitals of the same energy, electrons will first occupy all degenerate orbitals singly with parallel spin before any orbital is doubly occupied. Electrons will then pair up with an electron in one of the orbitals, with its spin opposite to the first electron in that orbital. This is because of the Pauli exclusion principle which states that no two electrons may have the same set of all four quantum numbers.

Atomic orbitals have the shapes they do because the wavefunctions of electrons in atoms naturally separate into :

$$\Psi_{n\ell m_\ell}(r,\theta,\phi) = R_{n\ell}(r)\, Y_{\ell,m_\ell}(\theta,\phi)$$

This separation explains why the spherical harmonics $Y_{\ell,m_\ell}(\theta,\phi)$ determine the angular behavior of the orbitals whereas the radial part only controls how far the electron extends from the nucleus. Orbital angular momentum operators are useful in order to calculate these spherical harmonics. Indeed, in spherical coordinates, orbital angular momentum operators are:

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi} \tag{28}$$

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right] \tag{29}$$

and their eigenfunctions are the spherical harmonics $Y_{\ell,m_\ell}(\theta,\phi)$. Orbital angular momentum has important physical consequences. It plays a role in selection rules since electric dipole transitions require $\Delta\ell = \pm 1$ and $\Delta m_\ell = 0, \pm 1$ [10]. It also contributes to the spin-orbit coupling interaction as we will see in the following section.

3.2 Spin-Orbit coupling

The magnetic moment and spin angular momentum for the free electron are related by $\vec{\mu_s} = g\frac{e}{2m}\vec{S}$, with g= -2 the Landé factor for electrons. This relation between the spin angular momentum and the magnetic moment gives rise to the spin–orbit interaction, which is important for describing the electronic structure of crystalline materials. Spin–orbit coupling is the interaction between an electron's intrinsic spin and the motion of the electron in an electric potential, producing an energy that depends on the relative orientation of spin and orbital motion. This potential typically corresponds to the Coulomb potential of a nucleus for an isolated atom or the periodic crystal potential in a crystalline solid. Physically, this links the orbital angular momentum L and the spin angular momentum S (see equation 30), creating energy splittings and modified selection rules. The Hamiltonian characterizing this interaction is described by the following equation [11] :

$$\hat{H}_{\mathrm{SO}}^{\mathrm{atomic}} = \frac{1}{2m^2c^2}\frac{1}{r}\frac{dV(r)}{dr}\hat{L}\cdot\hat{S} \tag{30}$$

with:

- m is the electron mass, c is the speed of light;
- $V(r)$ is the central potential;
- $\hat{L}$ is the orbital angular momentum operator;
- $\hat{S}$ is the spin angular momentum operator.

For an electron with orbital quantum number ℓ, SOC splits the $(2\ell+1)$ degenerate energy level into two multiplets with total angular momentum :

$$\hat{J} = \hat{L} + \hat{S}, \qquad j = \ell \pm \tfrac{1}{2} \tag{31}$$

The energy shift is proportional to the expectation value $\langle \hat{L} \cdot \hat{S} \rangle$ and using

$$\langle \hat{L} \cdot \hat{S} \rangle = \frac{1}{2} \left(\langle \hat{J}^2 \rangle - \langle \hat{L}^2 \rangle - \langle \hat{S}^2 \rangle \right) = \frac{\hbar^2}{2} \left(j(j+1) - l(l+1) - s(s+1) \right) \tag{32}$$

one obtains different energies for $j = \ell + \frac{1}{2}$ and $j = \ell - \frac{1}{2}$. This produces the fine structure in atomic spectra. As a result, the SOC weakly splits an orbital energy state into two states : one state with the spin aligned to the orbital field and one anti-aligned. In heavy atoms, the energy splitting can go as high as hundreds of meV to eV for core levels. Without SO interaction, energy bands in a crystal are also doubly degenerate due to spin (spin-up and spin-down). SO interaction splits these bands into spin-split levels, creating complex band structures. However, the strength of the SO interaction is not uniform throughout the crystal. Indeed, it depends strongly on the atomic number Z of the constituent elements and on the crystal lattice symmetry.

3.3 Crystal field theory

We now turn to crystal-field theory to understand how the electronic structure of the dopant ion emerges in the solid and how it gives rise to the energy-level diagrams used in laser operation.

It is a widely acknowledged truth that the complex interplay of the properties of the host lattice and the impurities determine the optical characteristics of dielectric crystals and in particular, their colors. This is due to the fact that the impurities produce localized electronic states within the band gap of the host. The optical transitions between these states, as well as between the host's electronic bands and the impurity energy levels, manifest themselves in the change of the color of crystal. Aluminum oxide Al_2O_3 is a well-known example of this kind. Its wide band gap of about 8.7eV makes it colorless and transparent in the visible part of the electromagnetic spectrum. However, if just 1% of the Al^{3+} ions are replaced with Cr^{3+} ions, the colorless aluminum oxide gets altered to ruby. The occurrence in the change of the crystal's color can be easily understood by referring to different electronic configurations of the dopant. Indeed, when a dopant is placed inside a crystal, the surrounding electric field from the crystal (called the crystal field) splits the dopant's energy levels into several sub-levels. As a result, the number and spacing of the energy levels that are split by the crystal field vary with respect to the dopant that we introduce in the crystal.

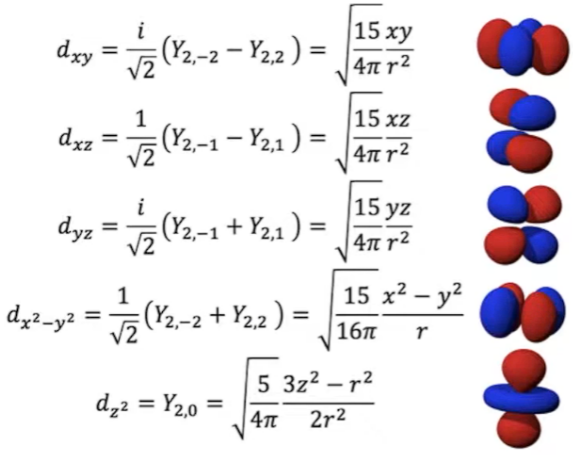


Figure 9: Cubic orbitals [12].

Let us now examine the crystal field felt by the electron orbitals in a real crystal. For this discussion, we will consider the closed packed crystal structure bcc, hcp and fcc adopted by transition metals. Those structure are not spherically symmetric, such that if we want to represent atomic orbitals in such crystals, spherical harmonics are not suitable anymore. We thus need to use new orbitals that respect the crystal symmetry of the lattice. This is done by recombining these spherical harmonics resulting in cubic orbitals (figure 9). Although cubic harmonics are better suited to describe cubic crystal structures, they do not necessarily remain degenerate in energy as we will see for the specific case of ruby, which will be generalized to sapphire afterwards.

3.3.1 Cr^{3+} in Octahedral Field (Ruby)

Let us consider an octahedral lattice made of Al_2O_3 where an ion Al^{3+} sits in the crystal central atom and is surrounded by 6 nearest neighbors O^{2-} forming an octahedron. In this case, the d_{xy}, d_{xz} and d_{yz} orbitals will have a lower energy than the d_{z^2} and $d_{x^2-y^2}$ orbitals. This is because the lobes of the d_{xy}, d_{xz} and d_{yz} orbitals extend in between the nearest neighbors while the lobes of the d_{z^2} and $d_{x^2-y^2}$ orbitals directly overlap with these nearest neighbors (figure 10). The former thus experiences much less electrostatic repulsion than the latter. The d_{xy}, d_{xz} and d_{yz} orbitals formed what is called the t_{2g} state which is a 3-fold degenerate state whereas the d_{z^2} and $d_{x^2-y^2}$ orbitals form the e_g state which is 2-fold degenerate. The energy difference between these 2 levels is called the crystal field splitting (≈ 2.320eV for ruby)[13] as it is governed by the crystal structure.

In addition to the octahedral nature of the lattice of ruby, this same structure is also distorted because of the doping. Indeed, when Cr^{3+} ions substitute Al^{3+} ions in the aluminum oxide crystal lattice, the surrounding O^{2-} ions compress slightly along the trigonal axis which cause further smaller splittings within the t_{2g} and e_g sets. Therefore, we still observe degeneracy of the d orbitals of chromium that split into different energy levels due to CFT but also the trigonal distortion further modifies the d-level splitting leading to an additional loss of degeneracy.

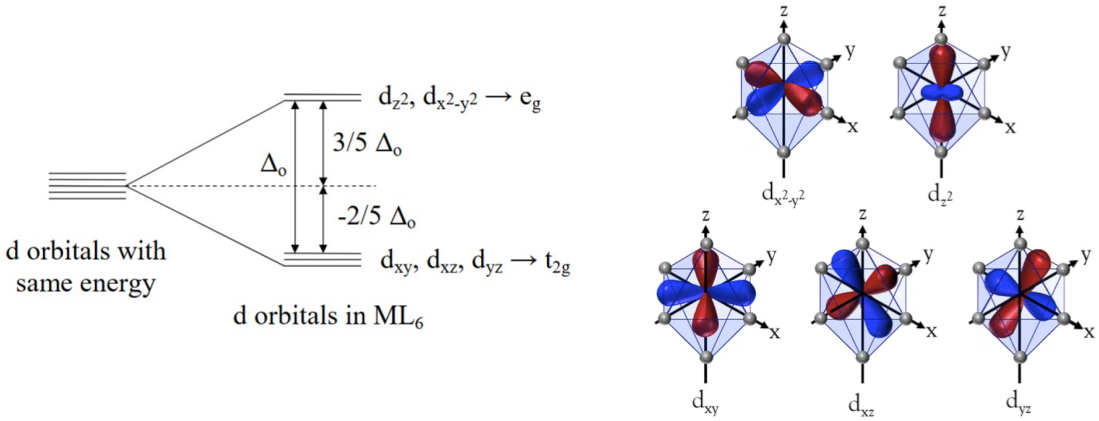


Figure 10: Five degenerate d orbitals of Cr^{3+} ion split into two sets t_{2g} and e_g separated by the crystal field splitting energy Δ_0 for an octahedral lattice. Note that all these doped levels are localized inside the band gap of the lattice [14] [15].

At the stage where no crystal field and no SOC splitting are included, the electronic configuration of Cr^{3+} is $3d^3$ so that its term symbol is ^4F for the ground state according to empirical Hund's rule. Then, taking into account the lattice of ruby, the free-ion term split into crystal-field multiplets as follows:

$$^4F \rightarrow {}^4A_2 + {}^4T_2 + {}^4T_1 \tag{33}$$

with $^4A_2$ the ground state, $^4T_2$ the first excited state and $^4T_1$ a higher excited state. Each wavefunction can be described by a Mulliken label (symmetry label). The labels used for terms in an octahedral ligand field are listed in the table below.

Mulliken Labels	**Meaning**
A or B	singly degenerate
E	doubly degenerate
T	triply degenerate
Subscripts	**Meaning**
1	symmetric to vertical mirror plane or perpendicular C_2 axis
2	anti-symmetric to vertical mirror plane or perpendicular C_2 axis

Within each crystal-field multiplet, SOC also produces small internal splittings sometimes called fine structure splitting. The famous R-lines (R_1 and R_2) appear in the case of ruby (figure 11). Indeed, SOC on Cr^{3+} splits the ^2E metastable state (level derived from the crystal splitting of the higher-energy ^2G free-ion term of chromium) into two states corresponding to the R_1 and R_2 transition. Because we must pump into both R_1 and R_2 manifolds when building a ruby-based laser, ruby exhibits two sharp laser lines. Therefore, these lines arise from the crystal field splitting of energy levels by the trigonal distortion of the lattice of ruby and spin–orbit coupling.

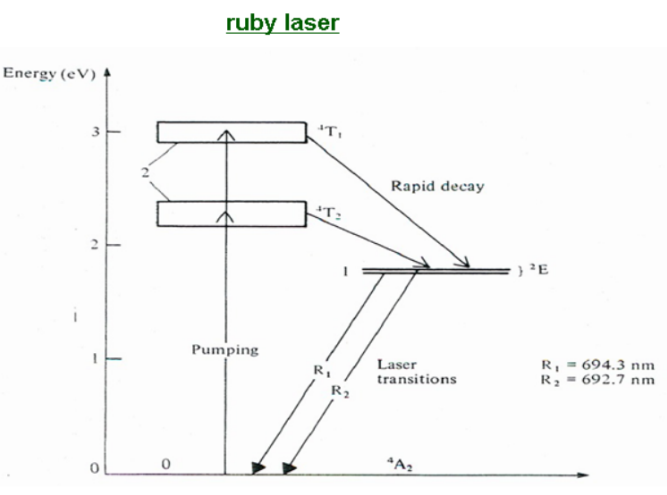


Figure 11: Energy level diagram and transitions for a 3-level ruby LASER [16].

The red color of ruby is a result of significant transmission and reflection in the red while the energy of the absorbed light, corresponding to the energy difference due to the d-orbital splitting levels, is in the blue and green regions of the visible spectrum. More specifically, Cr^{3+} absorbs green and blue light due to the following transitions :

$$^4A_2 \rightarrow {}^4T_2 \quad {}^4A_2 \rightarrow {}^4T_1 \tag{34}$$

Indeed, these transitions occur at wavelengths around 400-450nm and 550nm. From these levels, the electrons of the Cr^{3+} ions decay to the metastable energy level ^2E (lifetime of about 3ms[16]) in non-radiative transitions. The energy released in these non-radiative transitions is transferred to the crystal vibrations and changed into heat that must be removed away from the system. Then, the ^2E state splits into 2 sub-levels due to SOC, and the R_1 decay of the lower state at 694.3nm gives the red color of ruby and is the level used for population inversion in ruby lasers.

In ruby, the Cr–O distances are relatively short because of the constraints of the host lattice, which increases the d orbital–ligand interactions and makes the crystal field splitting relatively large. Ligands that cause a large splitting of the d-orbitals are known as strong-field ligands. In structures containing such ligands, placing electrons in the higher-energy orbitals becomes energetically unfavorable. As a result, electrons completely fill the lower-energy orbitals before occupying the upper ones, following the Aufbau principle and overriding Hund's rule. A greater crystal field strength in ruby implies a larger splitting of the Cr^{3+} energy levels, with subsequent blue shift of the absorption peaks in ruby when compared with those in emerald, which have longer bond distance.

3.3.2 Ti^{3+} in octahedral field (Sapphire)

Ti:sapphire has the same host lattice as ruby but Al^{3+} ions are substituted by Ti^{3+} cations instead of Cr^{3+}. Trivalent titanium has a $3d^1$ electron configuration and the free-ion electronic term for a single d electron is ^2D. The crystal field splits this term into $^2T_2$ and ^2E (crystal field splitting$\approx$ 2.2eV) and Jahn–Teller distortion [17] further splits these Stark levels into sublevels as shown in figure 12. Moreover, the $3d^1$ electron of Ti^{3+} strongly interacts with the surrounding oxygens and because different d orbitals deform the TiO_6 octahedron in different ways (Jahn–Teller effect), changing the electronic state forces a large lattice relaxation, producing strong vibronic coupling. This results in a dense manifold of sublevels separated by tens to hundreds of cm^{-1} thus yielding broad continuous absorption/emission bands rather than a few sharp lines (see figure 13). Therefore, the combination of crystal-field splitting and vibronic coupling is what gives Ti:sapphire its exceptionally broad wavelength tunability ranging between 660 nm up to 1050 nm. Adjustable built-in optical systems can be

used to tune the laser by selecting the wavelength of the laser output. Also, Ti:sapphire has a characteristic large absorption band between 400 and 650 nm (blue-green) due to ${}^2T_2 \rightarrow {}^2E$ transitions. This is the band used for pumping with blue/green lasers.

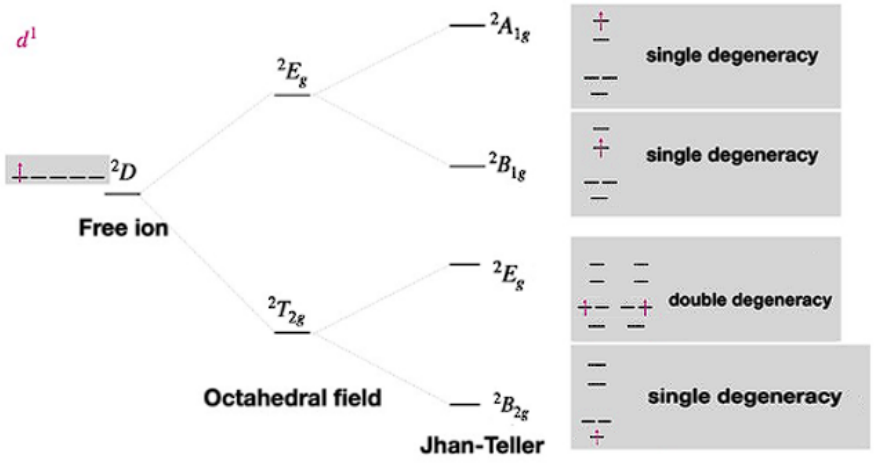


Figure 12: Splitting of d^1 by Jhan-Teller tetragonal distortion for an octahedral ligand field[18].

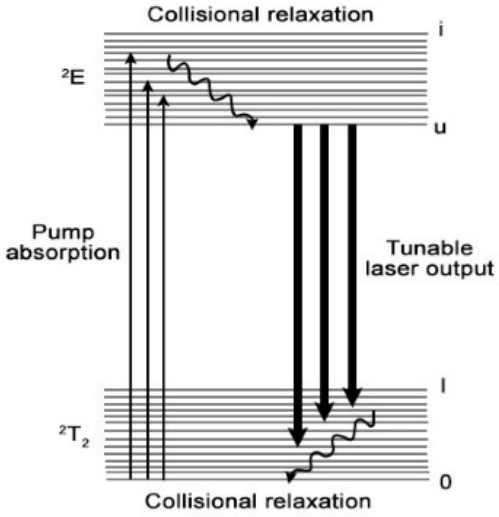


Figure 13: Energy level diagram and transitions of a 4-level Ti:Sapphire laser[19].

4 Gemstones: description and properties

Having established the quantum mechanical framework of crystal field theory, we can now understand how specific dopants like chromium or titanium behave when embedded in a crystal lattice. However, the optical properties of a laser depend not only on the ion but also on the host matrix itself. Before analyzing specific laser devices, we must first characterize the physical materials that serve as these hosts. In this section, we will define what constitutes a 'gemstone' in the context of functional materials, exploring their structural, mechanical, and thermal properties, as well as the fabrication methods required to synthesize them with laser-grade purity.

4.1 Structure and mechanical properties

In the study of gemstones, the mechanical properties of a gem arise from the arrangement of its constituent atoms within the crystal lattice. Many properties are investigated, among which the most notable are :

- **Cleavage**: occurring only in certain gemstones and along specific lattice directions, it is the formation of breaks due to the application of strong forces.[9]
- **Hardness**: often referring to the Mohs hardness scale, it is the tendency of a material to resist scratching.[20]
- **Fracture**: the formation of a break that does not follow a cleavage plane. Fractures can occur in any gemstone and in any direction.[21]

4.1.1 Sapphire and ruby

Ruby and sapphire are gem varieties of corundum, with chemical formula Al_2O_3. Corundum has a band gap of 6.044 eV, crystallizes in the trigonal system, and belongs to the $R\bar{3}c$ space group[22]. Its structure can be described as a hexagonal close-packed (HCP) arrangement of oxygen ions, with two-thirds of the octahedral interstitial sites occupied by aluminum cations. Constructing the full corundum unit cell requires stacking three HCP unit cells. When chromium (Cr^{3+}) or titanium (Ti^{3+}) ions substitute for aluminum, the local crystal structure is slightly distorted due to the larger ionic radii of Cr^{3+} and Ti^{3+} compared to Al^{3+}[22].

In Mohs Hardness scale, Corundum belongs to a class of minerals showing a value of 9, being second only to diamond. Moreover, ruby and sapphire not only cleavage along the cross sections of A-plane and M-plane, but also along the R-plane (which forms an angle of 57.6 degrees with the C-plane). These planes are well represented in figure 14.[23]

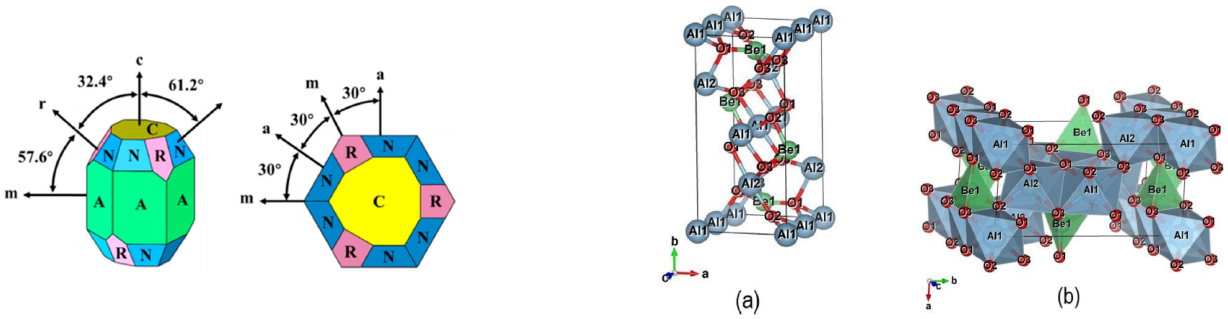


Figure 14: Corundum (left) [23] and chrysoberyl (right) lattice [24]. Corundum cleavage planes and angles are emphasized. Figure (a) and (b) of the chrysoberyl structure represent different views of the lattice itself, with particular emphasis on the location of the aluminium sites, Al1 and Al2. [24]

4.1.2 Alexandrite

Alexandrite ($BeAl_2O_4 : Cr^{3+}$) is a variety of the mineral chrysoberyl ($BeAl_2O_4$). Chrysoberyl crystallizes in the orthorhombic system, belongs to the *Pbnm* space group and has a band gap energy of 6.12 eV. Its unit cell consists in four molecules, with eight Al^{3+} ions (ionic radius 0.535 Å) occupying distorted octahedral sites, and four Be^{2+} ions (ionic radius 0.470 Å) located in distorted tetrahedral sites formed by oxygen ions arranged in planes perpendicular to the c axis[25].

Distortions of the crystal lattice produced by the surrounding oxygen atoms create two distinct aluminum sites, shown in Figure 14 : Al1, exhibiting inversion symmetry, and Al2, characterized by mirror symmetry. These sites can be substituted by transition metals such as iron, vanadium,

or—specifically in the case of alexandrite—chromium. Chrysoberyl has a Mohs hardness of 8.5, and its cleavage is distinct on {110}, imperfect on {010}, and poor on (001).[26][27]

4.2 Optical properties and the Alexandrite effect

In an undoped crystal, the electronic band structure does not allow absorption in the visible range, which makes the material appear colorless. When the band gap exceeds about 3.2 eV, photons in the visible spectrum do not have enough energy to promote electrons from the valence band to the conduction band, so no visible absorption or emission can occur.

The color of various gemstones arises from the presence of impurities within the crystal lattice, which replace the atoms naturally present in the structure. For example, in ruby, the characteristic red color is due to the partial replacement of Al^{3+} ions by chromium atoms[28][29].



Figure 15: Color change visible in an 8.07-ct alexandrite from Sri Lanka, from natural or fluorescent light (left) to incandescent light (right)[29].

In alexandrite, chromium ions preferentially substitute for aluminum in the Al2 sites (mirror symmetry), as this site is more spacious and distorted than the Al1 site (inversion symmetry). In fact, approximately 75-80% of the total chromium dopants occupy Al2 sites.

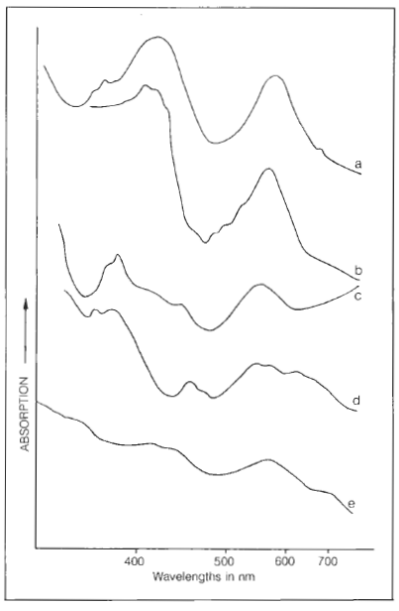


Figure 16: Absorption curves of gemstones presenting alexandrite effect: (a) alexandrite; (b) Mn-rich pyrope; (c) corundum; (d) spinel; and (e) fluorite[29].

From an optical standpoint, the Cr^{3+} ions in the Al2 site are the dominant contributors to the crystal's spectrum. Transitions in the Al1 site are dipole-forbidden due to inversion symmetry and are therefore much weaker. Consequently, it is the specific crystal field strength felt by the Cr^{3+} ions in the Al2 site that creates the broad absorption bands responsible for the 'alexandrite effect'—the material's ability to appear green in daylight and red under incandescent lighting (Figure 15).[29]
It can also occur in other crystal structures; for instance, a 20% Cr_2O_3 content renders corundum green in daylight and pink under incandescent light. The absorption curves of gemstones that showed alexandrite effect are reported in Figure 16: as it can be observed, the absorption spectra for all of these stones have transmission maxima in the red and blue-green regions, and an absorption maximum in the yellow region in common.[29][30]
Another key property is dispersion, which refers to the variation of the refractive index with wavelength. For gemstones used in lasers, it is essential that the refractive index remains as constant as possible across the relevant wavelengths, so that light can propagate efficiently through the material.

4.3 Synthetic gemstones and fabrication processes

Synthetic gemstones are indispensable in laser applications because they enable the production of crystals with high optical quality, purity, and uniformity, which cannot be reliably achieved with natural materials. Controlled growth eliminates inclusions and defects that would otherwise absorb or scatter light, thereby enhancing laser efficiency. Furthermore, the doping concentration can be precisely regulated, ensuring reproducible laser properties. Synthetic gemstones also provide superior thermal and mechanical stability, which is critical under intense pumping conditions. For these reasons, ruby, sapphire, and other synthetic crystals have become the standard in modern laser systems[31]. Two of the most used processes are covered in the following sections.

4.3.1 Czochralski method

The Czochralski technique is the primary method used to grow bulk single crystals of many electronic and optical materials. In this process, the starting material is placed into a cylindrical crucible and heated—typically by resistance or radio-frequency coils—until fully melted. A small seed crystal, only a few millimeters in diameter, is then lowered from above to touch the melt surface, where its tip is partially melted. At this contact point, a meniscus forms between the seed and the molten material. The seed is subsequently pulled upward at a controlled rate, often while rotating, prompting the melt to solidify at the interface and thus extending the crystal. A schematic view of this process is shown in Figure 17. Throughout growth, the crystal's geometry—especially its diameter—is regulated by adjusting parameters such as heater power, pulling speed, and rotation rate, usually through an automated diameter-control system. This system may monitor the meniscus profile or rely on weight measurements of either the crystal or the melt, depending on the material.

Crystal and crucible rotation are carefully tuned to manage heat and mass transport within the melt, which critically influences crystal quality by determining doping uniformity and the shape of the solid–liquid interface, and thereby the resulting thermal stresses. For electrically conductive melts, such as those used for semiconductor crystals, static or time-varying electromagnetic fields can be applied to further manipulate melt flow during the growth process[32][33].The technique is used to grow large ruby or sapphire crystals and rare earth garnets.

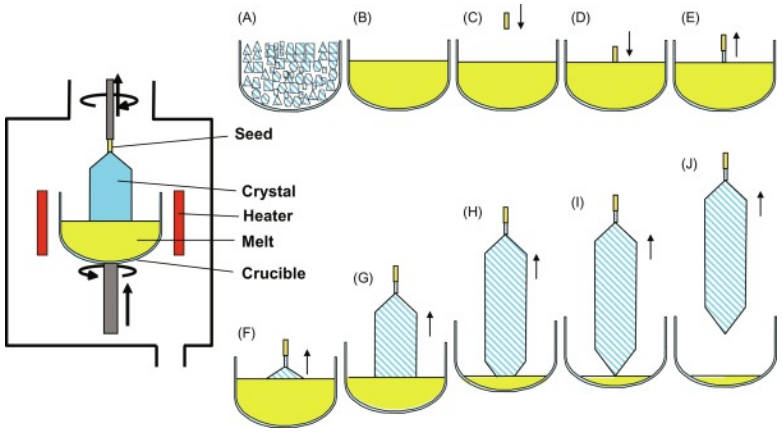


Figure 17: Schematic of the principle of the Czochralski method (left) and illustration of the different steps (a–j) of the Cz process for growing a Si crystal. (a) The polycrystalline feedstock is melted (b) in a crucible. (c, d) Seeding procedure: The seed crystal is dipped into the melt, followed by Dash necking (e), shouldering (f), cylindrical growth (g), growth of end cone (h), lift off (i), cooling down and removing of the crystal (j)[34].

4.3.2 Verneuil furnace

In this technique, crystal formation takes place in a specially designed furnace supplied with hydrogen and oxygen gases. A finely powdered starting material is introduced into the chamber through a narrow feed tube, where it mixes with oxygen before reaching an oxyhydrogen flame at approximately 2200 °C. As the powder traverses the flame, it melts into tiny droplets that descend onto a support rod. These droplets consolidate into a small sintered mound, which acts as the initial seed for crystal growth. Continuous deposition of molten droplets causes the support rod to be gradually lowered, enabling the formation of an elongated single crystal known as a boule-a term derived from the French word for "ball," referring to the shape of the early crystals. Typical boules reach lengths of about 100 mm with diameters between 15 and 20 mm, though their cross-section is not always perfectly circular. After growth, the boule is commonly split along its length to release internal stresses and prevent cracking.[35]

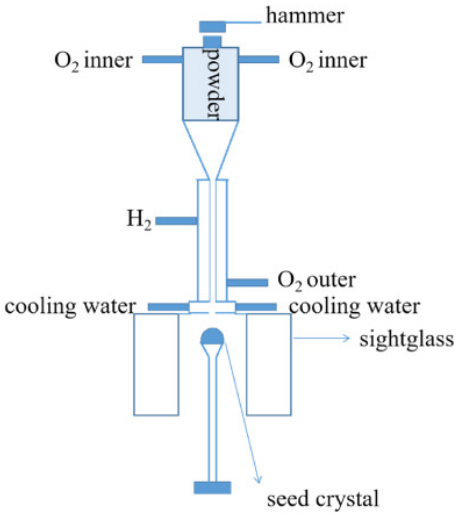


Figure 18: Representation of the Verneuil process[36].

5 Gemstones-based LASERs

Various fields benefit from the application of solid-state LASERs based on gemstones, such as medicine, research and industry. This section will explore the main applications for the 3 gemstones described earlier: ruby, sapphire and alexandrite. Table 1 gives an overview of the typical physical and optical characteristics of the 3 studied devices.

Characteristic	**Ruby**	**Ti:Sapphire**	**Alexandrite**
Active Ion	Cr^{3+}	Ti^{3+}	Cr^{3+}
Host Matrix	Al_2O_3	Al_2O_3	$BeAl_2O_4$
System Type	3-Level	4-Level	4-Level (vibronic)
Wavelength	694.3 nm (Fixed)	700–1000 nm (Tunable)	700–820 nm (Tunable)
Min Pulse Width	Nanoseconds (ns)	Femtoseconds (fs)	Microseconds (μs)
Key Application	Entertainment	Research / Ultrafast	Hair Removal

Table 1: Comparison of key physical and optical characteristics of gemstone lasers.

5.1 Ruby LASER

The ruby laser is the first solid-state LASER based on gemstone ever developed, in 1960. It was originally used in telemetry, especially in the military domain. Later, it was replaced with Nd:YAG LASER, much more efficient. Ruby lasers were also used in the medical field (for ophthalmology and dermatology) and for scientific research. But again, for the medical field, they were replaced by Nd:YAG LASER or alexandrite, wich we will talk about later [16].

Nowadays, ruby lasers are barely used in industrial applications because of their low repetition rate and efficiency [37], but are still used for applications where visible red light is required, such as in toys, laser pointers, tattoo removal, high speed photography or pulsed holography. These pulsed holograms

are used to study chemical reactions, since they often change the optical properties of solutions. Also, the generation of large holographic images is used in non-destructive testing laboratories, for example to detect defects in the design of aircraft tires.

Ruby lasers are well suited for this application because they can generate extremely intense pulses with durations shorter than a microsecond. Although holograms can also be produced using continuous-wave lasers, this requires much longer exposure times due to their lower beam intensity. Figure 19 illustrates a setup for recording and reconstructing a double-exposure hologram using a pulsed laser. In this arrangement, the highly coherent and monochromatic beam is divided into two paths by a beam splitter. One portion passes straight through the splitter and, with the aid of mirrors and lenses, illuminates the object to be recorded. The second portion, which is reflected by the beam splitter, is expanded and directed toward a high-resolution photographic plate without interacting with the object. This reference beam is then combined with the object-modulated light at the plate, where their interference pattern is captured by the recording medium. After photographic processing, this medium becomes the final hologram [38].

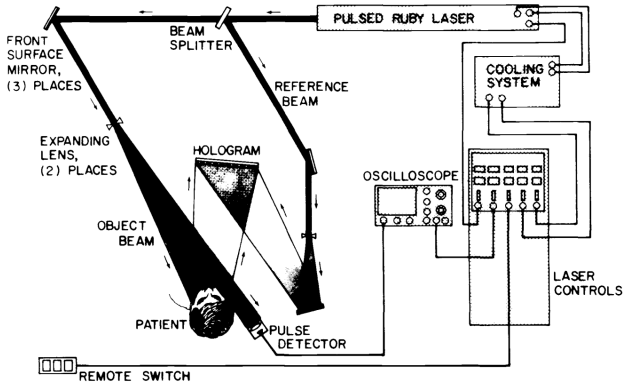


Figure 19: Schematic representation of the pulsed laser system for in vivo hologram interferometry[38]

5.2 Ti:Sapphire LASER

The Ti:sapphire laser is one of the most versatile solid-state lasers available nowadays, thanks to its exceptionally wide gain bandwidth (approximately 700–990 nm). This enables the generation of ultrashort optical pulses in the femtosecond regime. This type of LASER is thus used a lot in ultrafast science.

A major application of these LASERs is the generation of ultrafast X-ray and extreme-ultraviolet (XUV) radiation. When an intense femtosecond Ti:sapphire pulse interacts with atoms, it can ionize electrons and drive them in the strong optical field. Their recombination with the parent ion produces high-harmonic radiation, the basis of tabletop XUV and attosecond sources. Because the emitted photon energies depend on the instantaneous electric field of the driving pulse, this technique enables attosecond time-resolution and the tracking of electronic motion on its natural timescale.

The Ti:Sapphire LASERs are also very useful in biology, for the multiphoton microscopy (MPM), which is a noninvasive imaging technique. It provides high-resolution 3D imaging deep within thick, even in vivo, tissues. Since MPM relies upon a two-photon absorption, which is an nonlinear process, it requires a high peak power. Early MPM systems relied on femtosecond dye lasers: they provided the necessary peak power without causing thermal damage, but were complex and difficult to maintain. The development of turnkey, diode-pumped Ti:sapphire lasers made MPM practical and widely accessible.

Another application of the Ti:Saphire LASERs is the cold micromachining. It is because of their femtosecond pulses, which deliver energy faster than heat can diffuse, minimizing thermal damage and allowing precise cutting and drilling. [39]

5.3 Alexandrite LASER

The 755-nm Alexandrite laser is widely used in aesthetic and medical dermatology thanks to its ability to selectively target melanin through selective photothermolysis. Its primary application is long-term epilation, where its strong absorption by follicular melanin allows deep, efficient heating while limiting damage to surrounding tissues. Clinically, it achieves high hair-density reduction rates—around 85%, with relatively low rates of side effects.

Beyond aesthetics, it is also effective for treating medical conditions linked to abnormal hair growth, such as pseudofolliculitis barbae, acne keloidalis nuchae, and keratosis pilaris, where it reduces inflammation and follicular obstruction. Alexandrite lasers are also used to clear superficial pigmented lesions, including lentigines and freckles, by breaking down excess epidermal melanin. They have shown partial effectiveness in treating epidermal melasma, improving facial hyperpigmentation, though results are limited for dermal forms of the condition. Additionally, they are indicated for Becker's nevus, helping reduce both pigmentation and associated hair growth. [40]

6 Market and prospectives

6.1 Market analysis

The global laser market stands at USD 21.43 billion in 2025 and is forecast to reach USD 30.14 billion by 2030, translating to a 7.06% CAGR (Compound Annual Growth Rate). This expansion reflects rising deployment across precision micromachining, additive manufacturing, autonomous mobility, and next-generation display production. Ultrafast pulse sources that machine sub-10 nm semiconductor features and kW-class fiber systems that cut thicker metal sheets are now mainstream in high-volume factories. Government-funded photonics clusters accelerate ecosystem development in Asia-Pacific, while additive manufacturing lasers lower material waste in aerospace components and shorten production cycles. Asia-Pacific dominated with 46.9% share in 2024 and is anticipated to grow at 8.3% CAGR through 2030, buoyed by semiconductor and display manufacturing hubs. Fiber lasers commanded 41.8% revenue share of the global lasers market in 2024, while solid-state lasers are accelerating at a 9.3% CAGR to 2030 [41].

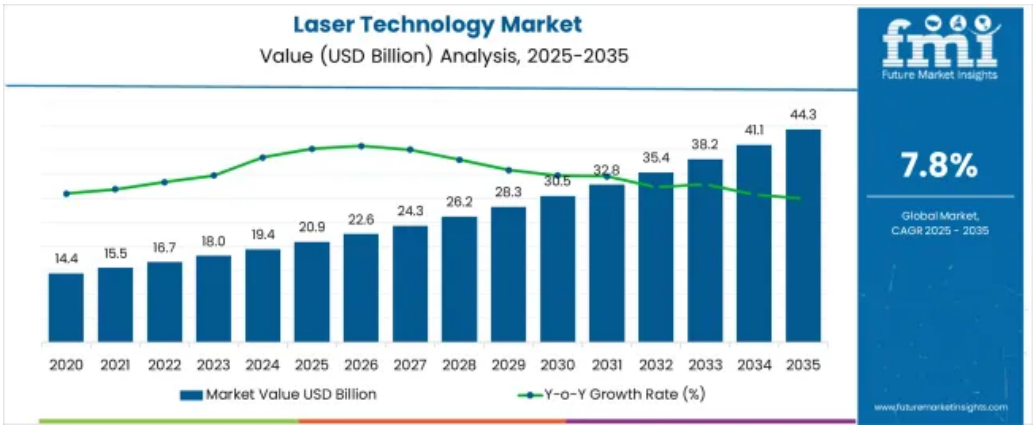


Figure 20: LASER technology market growth and projection[42]

Let us now focus on the solid-state (based on optical pumping) laser market, as it constitutes the core subject of this report. Figure 21 gives the value and growth of the solid-state laser market, as well as the share of each application. Regarding applications, solid-state lasers are used in industrial processing, medical industry, defense and aerospace, research and development, and entertainment.

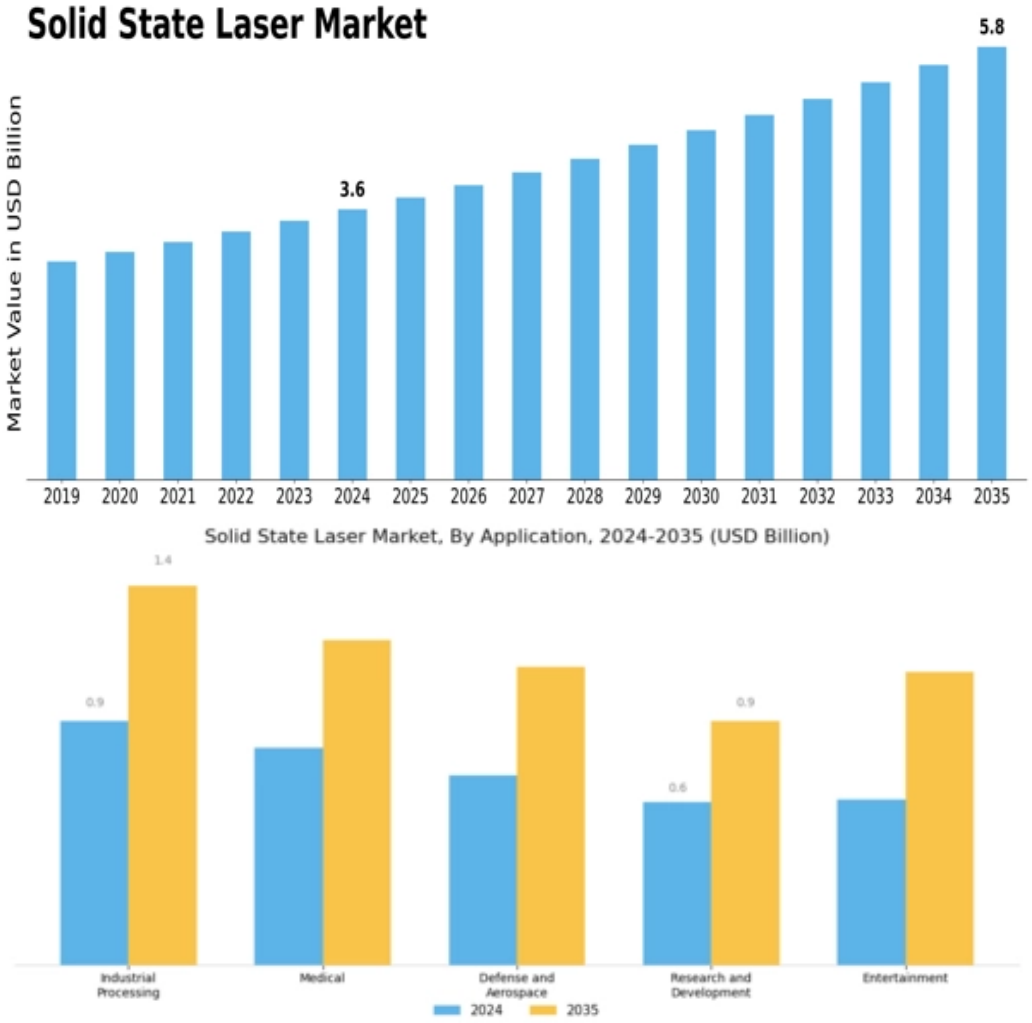


Figure 21: Global analysis of the Solid-State Laser market[43].

The solid-state laser market is sharply divided between ubiquitous industrial tools and high-value specialist devices. While this report focuses on **gemstone lasers** (Ruby, Alexandrite, Ti:Sapphire), it is important to contextualize their role. They are not mass-manufacturing tools; rather, they serve critical niches where their specific optical transitions are irreplaceable. **Ruby lasers** (694 nm), though largely historically significant, remain vital in dermatology for removing green and black tattoo pigments and in pulsed holography due to their high spectral coherence [44]. **Alexandrite lasers** (755 nm) have become the global gold standard in aesthetic medicine, specifically for hair removal, as their wavelength offers the optimal balance of melanin absorption and skin safety [45]. **Ti:Sapphire lasers** are the engine of modern photonics research; their extreme tunability and ability to generate femtosecond pulses make them indispensable for multiphoton microscopy and quantum computing research [46].

However, in terms of raw market volume, the solid-state sector is dominated by **Nd:YAG** (Neodymium-doped Yttrium Aluminum Garnet) lasers. Unlike the gemstones discussed here, Nd:YAG offers superior thermal conductivity and efficiency, making it the primary choice for heavy industrial applications like welding, cutting, and military range-finding [16]. Thus, while Nd:YAG powers the factory floor, gemstone lasers power the clinic and the laboratory. Figure 22 enlightens the market share of the

solid-state laser industry by materials.

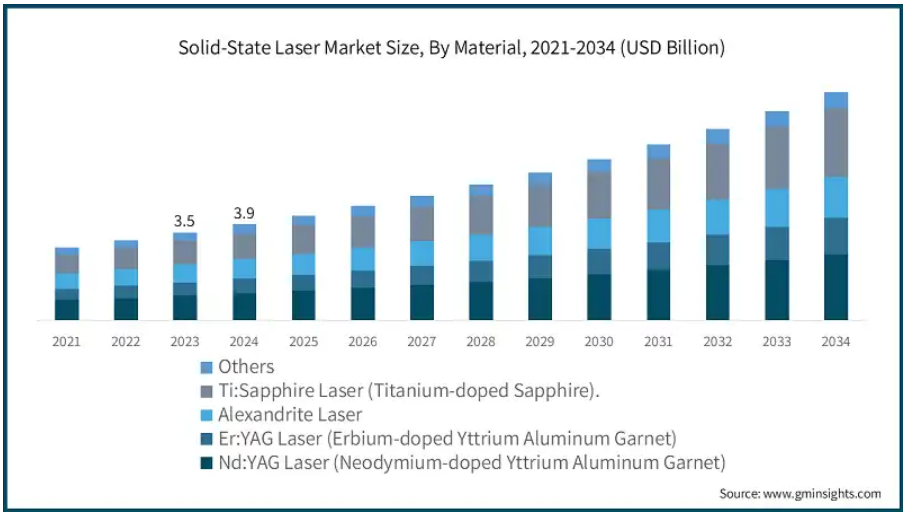


Figure 22: Solid-state laser market size by materials[47]

6.2 Future Prospectives: Miniaturization and Strategic Sovereignty

The future of gemstone-based lasers is shifting away from traditional massive tabletop systems toward integrated photonics and extreme-environment applications. Three key trends define the prospective landscape for the next decade.

The Chip-Scale Revolution (Ti:Sapphire on Insulator): For decades, the Ti:Sapphire laser has been the ultimate research tool but suffered from a major drawback: it is bulky, expensive, and requires complex alignment. A paradigm shift occurred in 2024 when researchers successfully demonstrated the first **chip-scale Ti:Sapphire laser** [48]. By bonding a thin layer of Titanium-doped Sapphire onto a silicon dioxide platform, they reduced the laser's footprint by four orders of magnitude (from typical cm^2 to μm^2). This breakthrough opens the door to mass-producible, low-cost ultrafast lasers for **quantum computing** and portable atomic clocks, potentially moving gemstone lasers from the optics lab to the consumer electronics sector.

Lasers in Space (Alexandrite LIDAR): While Ruby is declining, Alexandrite is finding a new frontier in orbit. Its ability to withstand high temperatures and its efficiency when pumped by diodes make it the ideal candidate for space-borne **LIDAR** (Light Detection and Ranging). Future satellite missions for climate monitoring aim to use compact Alexandrite lasers to measure wind speeds and atmospheric temperature profiles from space [49]. This represents a shift for gemstone lasers from purely medical uses to critical roles in Earth observation.

Supply Chain Sovereignty and Critical Minerals : As lasers become strategic military and industrial assets, the supply chain of the gemstones themselves faces scrutiny. Unlike mass-market semiconductors, laser crystals like Alexandrite ($BeAl_2O_4$) require **Beryllium**, a material classified as a Critical Raw Material (CRM) by both the EU and USA due to concentrated supply chains and toxicity [50]. Furthermore, the Czochralski growth process is extremely energy-intensive, requiring furnaces to run at >2000°C for weeks. Future market dominance will likely belong not just to those who can build the lasers, but to those who can secure the sovereign production of these high-purity synthetic crystals.

7 Conclusion and Outlook

This report has explored the intrinsic connection between gemstones and laser technology, a relationship that dates back to Theodore Maiman's first ruby laser in 1960. By combining solid-state physics, quantum mechanics, and materials science, we have answered the three fundamental questions proposed in our introduction.

Origin of Color and Tunability: We have shown that the color of gemstones is not a static property but a direct manifestation of **Crystal Field Theory**. The specific interaction between transition metal dopants (Cr^{3+}, Ti^{3+}) and the host crystal lattice (Al_2O_3, $BeAl_2O_4$) splits the d-orbital energy levels. Critically, this color *is* tunable. By selecting specific dopants or modifying the host matrix (as seen in the "Alexandrite effect"), we can engineer the energy gaps to absorb and emit light at desired wavelengths.

The Laser Principle: We established that a laser is more than just an amplifier; it is an oscillator that requires winning a "war" against absorption. Through the quantum mechanical description of light-matter interaction, we demonstrated that a **population inversion** is mathematically impossible in a simple two-level system. The solution lies in using the complex energy level structures of doped insulators—specifically 3-level (Ruby) and 4-level (Alexandrite/Sapphire) systems—activated by optical pumping inside a resonant cavity.

Why Gemstones? Finally, we explained that gemstones are not merely decorative but are high-performance engineering materials. Their use in high-power lasers is justified by two key factors: They provide the necessary metastable states and vibronic coupling (especially in Ti:Sapphire) to allow for efficient population inversion and extreme tunability, and the host matrices (Corundum, Chrysoberyl) offer exceptional hardness and thermal conductivity, allowing them to withstand the intense heat and optical fields of laser operation where other materials (like glass) would fail.

Looking forward, the sector is undergoing a paradigm shift. While gemstones have historically been associated with massive tabletop systems, 2024 marks the dawn of the chip-scale era. With the advent of Ti:sapphire-on-insulator technology, the future of these "precious" materials lies not in larger crystals, but in microscopic integration, promising to democratize quantum technology and medical diagnostics.

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References

[1] Wikipedia. Laser, 2025. Available: https://en.wikipedia.org/wiki/Laser [Accessed: 15/11/2025].

[2] T. H. Maiman. Stimulated optical radiation in ruby. *Nature*, 187:493–494, 1960.

[3] X. Gonze. LMAPR2014- Physics of Functional Materials lecture notes (long version), 2024.

[4] Matthieu Génévriez. Introduction to quantum optics. Lecture slides, 2025. Accessed: 2025-02-10.

[5] Carlucci Carlo. Notes of "fisica atomica". Personal notes, 2024.

[6] Simon Fraser University. Optical pumping. Lecture slides, 2025. Accessed: 2025-11-16.

[7] Rp photonics, 2025. Accessed: 2025-11-16.

[8] Antoine Passau, Diego Haya Enriquez, Augustin Vanderaa, and Nathan Wauthy. Gemstones and laser, 2020.

[9] International gem society, 2025. https://www.gemsociety.org/article/cleavage-gemology/ [Accessed: 2025-11-29].

[10] William M. MacDonald. Isotopic spin selection rule for electric dipole transitions. *Physical Review*, 98:60, 1955.

[11] N. Karim, M. I. Ul Haq, H. Schulz, T. Pfau, L. Santos, and T. Zibold. Experimental evidence for quantum many-body collapse in a bose–einstein condensate due to three-body losses. *Physical Review A*, 84:025602, 2011.

[12] Aurelien Manchon. Physique manchon — research. https://physiquemanchon.wixsite.com/research, 2020.

[13] M. O. J. Y. Hunault, Y. Harada, J. Miyawaki, J. Wang, A. Meijerink, F. M. F. de Groot, and M. M. van Schooneveld. Direct observation of Cr^{3+} 3d states in ruby: Toward experimental mechanistic evidence of metal chemistry. *The Journal of Physical Chemistry A*, 122(17):4399–4407, 2018. Detailed study of Crystal Field splitting in Ruby.

[14] Peter Atkins, Tina Overton, Jonathan Rourke, and Mark Weller. *Inorganic Chemistry*. Oxford University Press, 6 edition, 2014.

[15] Peter W. Atkins and Ronald Friedman. *Molecular Quantum Mechanics*. Oxford University Press, 5 edition, 2011.

[16] W. Koechner. *Solid-State Laser Engineering*. Springer, 2013. The standard reference for industrial Nd:YAG lasers.

[17] R. Janes and E. A. Moore. *Metal–Ligand Bonding*. The Open University, 2004.

[18] LibreTexts Contributors. Symmetry labels for split terms. https://chem.libretexts.org/Bookshelves/Inorganic_Chemistry/Inorganic_Chemistry_(LibreTexts)/11%3A_Coordination_Chemistry_III_-_Electronic_Spectra/11.03%3A_Electronic_Spectra_of_Coordination_Compounds/11.3.04%3A_Symmetry_labels_for_split_terms.

[19] Walter Koechner. *Solid-State Laser Engineering*. Springer Series in Optical Sciences. Springer New York, New York, NY, 6 edition, 2006.

[20] International gem society, 2025. https://www.gemsociety.org/article/hardness-and-wearability/ [Accessed: 2025-11-29].

[21] University of auckland, 2005. https://rocksminerals.flexiblelearning.auckland.ac.nz/minerals/fracture.html [Accessed: 2025-11-29].

[22] Pascal Jacques. Notes from the course"science and engineering of metals and ceramics". Lecture slides,Chap4, 2025.

[23] S. Huang et al. Fracture behavior of single-crystal sapphire in different crystal orientations., 2021. Available: https://doi.org/10.3390/cryst11080930 [Accessed:2025/11/20].

[24] Jaspreet Singh et al. High-pressure structural, lattice dynamics, and electronic properties of beryllium aluminate studied from first-principles theory., 2021. Available: https://doi.org/10.1016/j.mtcomm.2020.101801. [Accessed:2025/11/20].

[25] A.O. Silva et al. Stimulated luminescence properties of natural alexandrite in response to x-ray irradiation., 2024. Available: https://doi.org/10.1016/j.jlumin.2024.120493 [Accessed:2025/11/20].

[26] A. Bordage et al. Site partitioning of cr3+ in the trichroic alexandrite beal2o4:cr3+ crystal: contribution from x-ray absorption spectroscopy., 2012. Available: https://doi.org/10.1088/0953-8984/24/22/225401 [Accessed:2025/11/21].

[27] O. Rybnikova et al. Characterization of chrysoberyl and its gemmological varieties by raman spectroscopy., 2023. Available: https://doi.org/10.1002/jrs.6566 [Accessed:2025/11/21].

[28] Ruby fluorescence activation spectra, 2020. Available: https://physics.stackexchange.com/questions/364266/ruby-fluorescence-activation-spectra [Accessed: 2025-11-23].

[29] E. J. Gübelin and Karl Schmetzer. Gemstones with alexandrite effect. *Gems & Gemology*, 18:197–203, 1982. Available: https://api.semanticscholar.org/CorpusID:54512468 [Accessed:2025/11/24].

[30] Yan Liu et al. The "alexandrite effect" in gemstones, 1994. Available: https://doi.org/10.1002/col.5080190306 [Accessed:2025/11/24].

[31] Understanding gem synthetics, treatments, and imitations, 2024. Available: https://www.gemsociety.org/article/understanding-gem-synthetics-treatments-imitations-part-4-synthetic-gemstone-guide/ [Accessed: 2025-11-29].

[32] G. Müller and J. Friedrich. Crystal growth, bulk: Methods. In Franco Bassani, Gerald L. Liedl, and Peter Wyder, editors, *Encyclopedia of Condensed Matter Physics*, pages 262–274. Elsevier, Oxford, 2005.

[33] J. Friedrich. Methods for bulk growth of inorganic crystals. In Tapash Chakraborty, editor, *Encyclopedia of Condensed Matter Physics (Second Edition)*, pages 190–207. Academic Press, Oxford, second edition edition, 2024.

[34] Friedrich et al. Czochralski growth of silicon crystals. In Tapash Chakraborty, editor, *Handbook of Crystal growth: Bulk Crystal growth*, pages 47–61. Elsevier Science, Amsterdam, 2014.

[35] Crystal growth. Available:https://www.alineason.com/en/knowhow/crystal-growth/ [Accessed: 2025-11-29].

[36] Lei Wang et al. Verneuil growth of tio2 (rutile) crystals of large size and low dislocation density at low gas flow rate. *Ceramics International*, 2022. Available: https://doi.org/10.1016/j.ceramint.2022.06.297.

[37] Azo optics. Ruby laser – history, properties and applications, 2013. Available: https://www.azooptics.com/Article.aspx?ArticleID=466 [Accessed: 06/12/2025].

[38] Ryszard J. Pryputniewicz. *Pulsed Laser Holography in Studies of Bone Motions and Deformations*. Worcester Polytechnic Institute, 1985.

[39] Julien Klein. The ti:sapphire laser, 2010. Available: https://www.photonics.com/Articles/The-TiSapphire-Laser/a42950 [Accessed: 05/12/2025].

[40] Candice Smarrito-Menozzi. Laser alexandrite 755 nm en dermatologie esthétique, 2017. Available: https://rivieraclinicacademy.ch/laser-alexandrite/ [Accessed: 05/12/2025].

[41] Mordor Intelligence. Lasers market size & share analysis - growth trends & forecasts (2025 - 2030), 2025. Available: https://www.mordorintelligence.com/industry-reports/lasers-market [Accessed: 2025-11-30].

[42] Future Market Insights. Laser technology market outlook, trends, and forecasts, 2025. Available: https://www.futuremarketinsights.com/reports/laser-technology-market [Accessed: 2025-11-30].

[43] Market Research Future. Solid-state laser market size, share & industry trends analysis report, 2025. Available: https://www.marketresearchfuture.com/reports/solid-state-laser-market-24422 [Accessed: 2025-11-30].

[44] J. Hecht. A short history of laser development. *Applied Optics*, 49(25):F99–F122, 2010.

[45] K. Nouri. *Lasers in Dermatology and Medicine*. Springer, 2018.

[46] Newport Corporation. The ti:sapphire laser: Applications in research and industry, 2024. Available: https://www.newport.com/n/the-tisapphire-laser [Accessed: 2025-11-30].

[47] Global Market Insights. Solid-state laser market size, share & trends forecast – 2034, 2024. Available: https://www.gminsights.com/industry-analysis/solid-state-laser-market [Accessed: 2025-11-30].

[48] Joshua Yang et al. Titanium:sapphire-on-insulator integrated lasers. *Nature*, 630:853–859, 2024. A major breakthrough shrinking gemstone lasers to chip size.

[49] Electro Optics. Compact alexandrite lasers to enable long-range lidar, 2024. Available: https://www.electrooptics.com [Accessed: 2025-11-30].

[50] RAND Corporation. Critical mineral suppliers and dependencies in the u.s. security context, 2024. Analysis of Beryllium and strategic laser materials.